WEEK 3

1) What is the primary objective of the Leader Election problem in distributed systems?
A. To elect multiple coordinators for load balancing
B. To ensure every process becomes a leader at least once
**C. To ensure exactly one non-faulty process is chosen as leader**
D. To minimize message delays in the network

2) Leader election is categorized as which type of problem in distributed systems?
A. Deadlock detection problem
**B. Symmetry-breaking problem**
C. Mutual exclusion problem
D. Load balancing problem

3) Which of the following is a valid use of a leader in a distributed system?
A. Encrypting all inter-process messages
**B. Coordinating activities such as spanning tree construction**
C. Increasing physical network bandwidth
D. Eliminating message loss

4) Why is leader election impossible in anonymous rings, even if the ring size is known?
A. Processes may crash arbitrarily
B. Messages are delivered out of order
**C. All processors start in identical states and remain indistinguishable**
D. The ring is unidirectional

5) Which assumption enables leader election to become possible in ring networks?
A. Synchronous message delivery
B. Knowledge of ring size
**C. Unique process identifiers**
D. Bidirectional communication

6) The LCR algorithm elects which process as the leader?
A. Process with smallest identifier
B. Process closest to initiator
**C. Process with largest identifier**
D. Process that sends the first message

7) What is the worst-case message complexity of the LCR algorithm?
A. O(n)
B. O(n log n)
C. O(log n)
**D. $O(n^2)$**

8) In the LCR algorithm, what happens when a process receives its own identifier back?
A. It forwards the identifier again
**B. It declares itself leader**
C. It discards the message
D. It starts a new election

9) What is the overall message complexity of the Hirschberg–Sinclair algorithm?
A. $\Theta(n^2)$
B. $\Theta(n)$
C. $\Theta(\log n)$
**D. $\Theta(n \log n)$**

10) In the Ring Election protocol, messages are forwarded in which direction?
A. Randomly
B. Both clockwise and counter-clockwise
**C. Clockwise only**
D. Based on process identifier

11) What is the worst-case message complexity of the Ring Election protocol (without failures)?
A. N
B. 2N
**C. 3N − 1**
D. $N^2$

12) ZooKeeper is best described as:
A. A distributed database for big data storage
**B. A centralized locking kernel for coordination**
C. A message-passing middleware
D. A distributed computation engine

13) Which ZooKeeper design goal ensures updates are processed in strict order?
A. Replicated
B. Fast
**C. Ordered**
D. Durable

14) Why does ZooKeeper require a majority of servers to be available?
A. To reduce latency
B. To avoid split-brain and ensure consistency
**C. To increase throughput**
D. To reduce storage overhead

15) Which type of znode is automatically deleted when the client session ends?
A. Persistent

B. Sequential
C. Replicated
**D. Ephemeral**